25900360 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT202 - SENSORS AND ACTUATORS IN INDUSTRY AUTOMATION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. The gain of a standard 3 op-amp instrumentation amplifier is adjujested by [U] / [CO1]

   a). A diode in feedback loop

   b). Supply voltage

   c). A variable capacitor in output stage

   d). One resistor connected between input amplifiers

2. Indicate the working principle of shunt resistor current sensor. [U] / [CO1]

   a). Ohms law (V=I*R)

   b). Capacitance variation

   c). Electromagnetic induction

   d). Peizoelectric variation

3. In automobiles, temperature sensors are [U] / [CO1]

a). Measuring tire pressure

b). Monitoring engine coolant temperature

c). Airbag deployement

d). Measuring vibrations

4. The environmental factors that affect the accuracy of Ultrasonic sensor is [U] / [CO1]

a). Magnetic field

b). Electric field

c). Temperatur and Humidity

d). Light intensity

5. The factors that affect the performance of IR sensor are [U] / [CO1]

a). Dust and smoke

b). Obstructions in the path

c). Strong sunlight

d). All of the above

6. Identify the ground (negative supply) pin of LM358. [K] / [CO2]

a). Pin 4 b). Pin 5

c). Pin 7 d). Pin 8

7. Select the Wheatstone bridge configuration that provides maximum sensitivity for strain measurement. [K] / [CO2]

a). Quarter bridge b). Half bridge

c). Full bridge (four active gauges) d). Open bridge

8. Describe the main advantage of SAR ADC. [U] / [CO2]

a). Very high resolution but slow

b). Very fast but inaccurate

c). Fixed conversion time with good speed and accuracy

d). Cheaper than all other ADCs

9. Select the correct passive transducer. [K] / [CO2]

a). Thermocouple b). Photovoltaic cell

c). Strain gauge d). Piezoelectric crystal

10. Explain on what the sensitivity of an LVDT mainly depends. [U] / [CO2]

a). Core material b). Number of turns on secondary coils

c). Frequency of excitation d). All of the above

11. Interfacing an actuator with a drive circuit typically involve [An] / [CO3]

a).Only connecting wires b).Matching electrical signal and power requirements

c).Installing software drivers d).Physical mounting only

12. What is the main component of a relay? [An] / [CO3]

a).Capacitor b).Diode

c).Coil d).Transistor

13. In solar energy setups, linear actuators are used to: [An] / [CO3]

a).Rotate solar panels randomly b).Tilt panels to track the sun efficiently across the sky

c).Cool down panels using water spray d).Shift panels in random directions

14. Actuators can generate which types of motion? [An] / [CO3]

a).Only linear b).Only rotary

c).Both linear and rotary d).Neither

15. Electric linear actuators convert electrical signals to: [An] / [CO3]

a).Pressure waves b).Linear physical movement

c).Voltage output d).Sound waves

16. Which of the following is the lowest level in the industrial automation hierarchy? [U] / [CO4]

a). Enterprise level b). Control level

c). Field level d). Supervisory level

17. What does PWM stand for in fan speed control? [K] / [CO4]

a). Pulse Width Modulation b). Power Wave Management

c). Pulse Weight Monitoring d). Programmable Width Module

18. Which library is commonly used in Arduino IDE to communicate with ThingSpeak? [U] / [CO4]

a). Servo.h b). ThingSpeak.h

c). Wire.h d). SPI.h

19. What does SCADA stand for? [K] / [CO4]

a). Supervisory Control And Data Acquisition b). Sensor Control And Device Automation

c). Software Control And Data Application d). System Configuration and Device Allocation

20. Which platform is often used to visualize sensor data from IoT water pump systems? [U] / [CO4]

a). WhatsApp b). ThingSpeak

c). Microsoft Word d). Chrome

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Explain industrial sensors like temperature or pressure sensor with example. [U] / [CO1]

22. List any five applications of temperature sensors in industry and daily life . [U] / [CO1]

23. Compare the advantages and limitations of R–2R ladder DAC. [U] / [CO2]

24. Differentiate between mechanical switches and touch switches. [U] / [CO2]

25. Explain the concept of step angle. [An] / [CO3]

26. .Describe the working principle of a solenoid [An] / [CO3]

27. Explain the concept of smart sensors in the context of IoT. How do they differ from traditional sensors? [U] / [CO4]

28. What is an LDR sensor, and how does it work for measuring light intensity? [K] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Explain the application of IR sensor in proximity detection and obstacle avoidance with neat diagram. [U] / [CO1]

30. Describe the different types of microphones with neat sketches. [K] / [CO2]

31. Explain the fundamental working principle of a servo motor, emphasizing its closed-loop control and feedback mechanisms. [An] / [CO3]

32. Discuss how smart sensors can be implemented using ESP32 and ThingSpeak in an IoT system. [K] / [CO4]

[10x2 = 20]